

A Survey of Stock Market Prediction - Based on Machine Learning Techniques

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Abstract—Stock market Prediction has been a topic of attention for numerous researchers since its beginning. Often traditional statistical methods get conflict to grab the complex, non-linear patterns in stock market data. Due to various factors predicting stock prices is challenging, including a company's financial performance, present assets, politics, general economic conditions, politics, unanticipated occurrence, global situation, etc. The fluctuating behavior of the inventory place market makes it difficult to expect stock prices, as they constantly change. This paper comprehensively reviews machine and deep learning techniques used to predict the stock market and highlights the proper techniques for predicting inventory cost. This paper also covers the performance metrics for measuring the performance of prediction models, research gaps, and issues found in previous work.

Keywords— *Stock market, machine-learning, prediction, deep-learning.*

I. INTRODUCTION

To make money, investors purchase or sell the stocks of companies indexed on the stock exchange. It's a crucial sign of a nation's economic well-being because it reflects the development of companies and how businesses are growing [1]. Shares are marketed, converted, and disturbed in the stock exchange. It is regarded as an important network for large companies to increase growth property from stockholders [2]. Several factors affect the stock market, making it sensitive, including politics, society, and the economy, among others [3]. Because it depends on financial, political, environmental, and a variety of internal and external factors along with past stock data, the stock market is highly volatile in the world [4]. To get high returns, investors put their amount in the bond market [5]. The growth of a country's financial market is a major aspect of the global economic situation and allows economists and financial experts to estimate the country's present economic health. The stock market stands out key driving force in contrast to other financial markets [6]. Changes in local and worldwide market economic schemes may cause growth to the stock market crisis [7]. The compound utilization of different causes and the notable efforts of single aspects, political, encompassing market, technological, and economic aspects, furthermore affect investor actions, will all lead to modifications in stock cost [8]. Accurate prediction of stock price advises investors for trading decisions, resulting in significant benefits [9]. The attempt designed to estimate a stock's future value, market sector, or

even the total market is called Stock Market Prediction price. [10].

A technique to reduce the risk investment or a prediction model should be appropriate in the stock market to ascertain the path of the market movement. Technical and fundamental are two primary approaches for modeling the stock price. In fundamental analysis, the future of stock prices depends on companies' financial analyses or sectors like interest rates, inflation, international relations, listed business finances industrial policy, and exchange, etc. which impact the stocks. The motive of fundamental analysis is to purchase an asset as soon as the cost is less and resell it at a profit by figuring out how much the asset's true value differs from the price in the market. The technical analysis makes use of past data to forecast future values by considering history tends to repeat itself [11].

Many researchers use machine and deep learning techniques to accurately expect the cost of stocks to achieve high returns in the financial center [5]. Due to its volatile nature, and the complexity of financial markets rise, the use of a machine-learning approach to foresee the stock in trade has garnered significant attention [12]. In finance, the most significant and significant use of machine learning technology is to forecast stock trends [13]. Initially, regression traditional methods are utilized to forecast stock movement. Data in stock can be classified as modular time series data; hence non-linear machine-learning techniques have also been applied. Support vector machine (SVM) algorithms and artificial neural network (ANN) algorithms are habitually used to foresee stock movements and stock price indices[14].

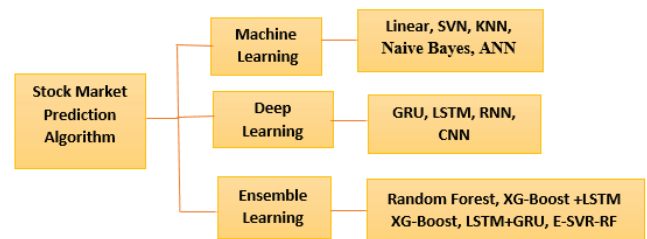


Fig. 1: Stock-Market Prediction Techniques

The majority Supervised classification algorithm is used in the analysis of stock market analysis including SVM. k-nearest neighbor (KNN), naive bayes, random forests, and decision trees. Both linear and multiple regression are used in regression

models. The two deep learning (DL) techniques that were often utilized in the prior study are long short-term memory (LSTM) and recurrent neural networks (RNN) [15]. The majority of researchers utilized Naive Bayes. The algorithm auto-regressive integrated moving average (ARIMA) forecasts stock value using time series forecasting. The utility of deep learning algorithms is growing in inventory market forecasting. For example, Long et al. employed a deep neural network with transaction records and considering public market data to evaluate stock patterns, with bidirectional LSTM achieving remarkable performance. Other studies used RNN, convolutional neural network (CNN), and LSTM to expect stock trends accurately. These models justify how deep learning can significantly improve prediction performance [16].

To measure the performance results of experiments, compare the real with the predicted model values. Evaluation metrics also called performance evaluations are accustomed to check the predictive model's performance which can be measured by accuracy and closeness parameters. The performance evaluations for accuracy are recall, F-measure, precision, and accuracy, for closeness performance evaluations are mean absolute percentage error (MSE), root mean square error (RMSE), mean absolute error (MAE), and R-square (R2) [16].

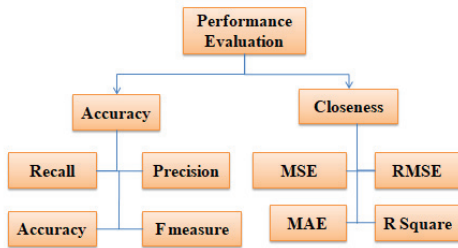


Fig. 2: Performance Evaluation

The structure of this paper is as follows: Section II describes the Related Work on different techniques and datasets used, it provides an overview of, many studies conducted in the field categorized under traditional ML and DL Techniques, Learning, Graph-based, and Time Series analysis approaches, respectively. Section III explores the major findings. Section IV discusses the conclusions from the study.

II. RELATED WORK

We examine the studies on previously initiated stock price prediction, and the analysis table as displayed in Table I. Aida [17] proposed the hybrid GARCH-LSTM model with 10 input nodes, including 2 hidden layers, and 1 output node having the best performance, with the lowest MAE, heteroscedasticity mean absolute errors (HMAE), RMSE, and mean absolute percentage error (MAPE) made a comparison with single LSTM model. Burak Gülmez [1] used the LSTM-ARO model which is comprised of three LSTM layers with variable numbers of neurons, a dense, a dropout, and an input-output layer were all employed together to see a stock price for the future by considering past data of stock price from the DJIA index, with 20% designated for testing and training for 80% of the data. Muhammad et al. [2] recommended a blended method that

blends the LSTM with a novel iteration of the empirical-mode-decomposition (EMD) technique to foresee the day-to-day closing prices of the Pakistan stock exchange (PSX) by using a dataset of KSE-100 index, outperforming other hybrid models and single models expressed by considering statistical measures namely: MAE, RMSE, and MAPE. The average accuracy of LSTM is 97.31 % whereas ARIMA is 82.73%, which declares that LSTM was better than ARIMA. The ARIMA model took significantly longer to process the data, but it could produce better results if faster processing devices were used or if the dataset had fewer missing values. Both LSTM and ARIMA models generated comparatively accurate results and were possibly beneficial for stock traders. The problem identified is data gathered from different nations must be measured with the suggested method. The flaw is, that the comparisons should be made with other ML models like SVM, XGBoost, ANN, and ANN – SVM models.

Warda [18] suggested a novel system called SMP-DL to predict stock closing price and also designed a mobile application called Smart Trading Platform (STP) that allows anyone to invest on a stock exchange, with the BiGRU-LSTM hybrid model outperforming LSTM, and GRU models to foresee closing prices of market. Akshat [19] used PSO ensembled with optimized LSTM and GRU models to forecast changes in the Nifty 50 index's price. The ensemble approach increased the precision of the predictions from 0.2882 to 0.5485. Jimmy [20] proposed- stock sequence array convolutionalLSTM (SACLSTM), for predicting the rise and fall movement using Ten American and Taiwanese stocks. Various news collections, and historical data, including options, and futures, involve the SACLSTM algorithm for stock prediction. When compared directly with earlier methods, the suggested algorithm performance is superior. Chin [9] utilizes the LSTM algorithm to improve investment success and enhance stock price forecasting by using historical stock data for one hundred stocks listed on Bursa Malaysia, including volume, adjusted closing price, high, volume, low, and open prices.

Guangyu [21] proposed a multi-source data-driven stock price prediction model called the MS-SSA-LSTM, which combines multiple sources of data and SSA-optimized LSTM and demonstrates high reliability, accuracy, and versatile in predicting stock prices, especially for the highly volatile. The proposed MS-SSA-LSTM combines historical stock data and investor sentiment to improve forecasting accuracy. The sentiment analysis only divides emotions into positive and negative. Somenath [24] used ANN and CNN, to foresee the cost of the financial market by demonstrating effectiveness even during the COVID-19 pandemic, the author used 2-D histograms generated data from the financial center as input to a convolutional neural network to divine the following day's stock prices. A feedforward and backpropagation neural network achieved 97.66% accuracy, and CNN using 2D histograms achieved 98.92% accuracy.

Yu [22] proposed the MCC model to foresee the stock prices backward and forward movement for the next day trading. The author used 31 quantitative indicators to represent the target stocks and GCN to model the relation between related stocks and target stocks.

TABLE I. ALGORITHMS USED, DATASET, EVALUATION METRICS USED, PURPOSE, ADVANTAGES, AND DISADVANTAGES

Reference	Authors / Published Date	Algorithms / Technique Used	Dataset	Evaluation metrics used	Purpose	Advantages	Disadvantages
17	Aida et al. (2024)	GARCH-LSTM	KLSE	RMSE, MAE, MAPE, HMAE, ARCH-LM test	To forecast the KLSE stock price index's accuracy.	Outperformed the single LSTM model in forecasting the KLSE stock price index, achieving 15% improvements in MAE and 13% in RMSE.	It is time-consuming and costly, and it takes a lot of trial and error to find appropriate parameters.
01	Burak Gulmez (2023)	LSTM-ARO	DJIA index data	MSE, R2 score, MAE, MAPE	To predict stock market prices.	The most effective model for fitting the data.	More complex
02	Muhammad et al. (2023)	EMD and LSTM	KSE-100	RMSE, MAE, MAPE	To predict the stock market's closing price.	More accuracy than the typical classical time series in predicting the stock price and its up-and-down movement. Achieved MAE value: 234.340 and MAPE value: 0.578	Complexity to reduce useless information.
18	Warda et al. (2023)	LSTM and BiGRU	IBM, AAPL, and GOOG Alphabet stocks	MAE, MSE, RMSE, R-squared	To predict the stock market's closing price.	Real-time trading, cost-effective, instant access to market data, and real-time notifications. -Achieved values: RMSE: 0.0831, MSE: 0.2883, MAE: 0.2099, and R2: 0.9948.	It performs better only when applying some datasets.
19	Akshat et al., (2023)	PSO + LSTM and LSTM + GRU	NSE	accuracy, recall, precision, F1-Score	Stock price movement-accuracy and precision	Achieved a high accuracy rate: 57.72%, F1-Score: 0.5042, precision: 0.5485, and recall: 0.531	Lack of sentiment analysis and technical indicators
20	Jimmy et al., (2023)	LSTM and CNN	U.S.A. and Taiwan.	accuracy	To foresee the fluctuations in the stock market	Compared to others (SVM, NN, CNNpre) SACLSTM is better performed in stock rise and fall predictions, enhancing overall accuracy when compared to historical data on individual prediction	Complexity to reduce useless information.
9	Chin et al., (2023)	LSTM	100 Malaysia n stocks	Prediction accuracy, Sharpe ratio, maximum drawdown, cumulative return	To improve stock levy prediction and investment performance.	-accuracy of prediction investment returns while minimizing risks. -Average prediction accuracy of 57% and a cumulative return of 29% across 100 stocks	- Performed well when stock is bullish than in a bear market. -Did not cover other metrics like RMSE, MAPE, etc.
21	Guangyu et al. (2023)	MS - SSA - LSTM	Chinese stock market.	MSE, MAE, MAPE, R ² score	To predict stock closing prices and to improve forecasting accuracy.	On average MS-SSA-LSTM is improved by 10.74% better than that of ordinary LSTM.	The model doesn't take into account other variables beyond sentiment and historical stock data.
22	Yu et al. (2022)	GCN, BiLSTM	Chinese stock market	ACC and MCC	To predict- stock price movement for next trading day.	A higher Sharpe Ratio in contrast to the state-of-art baselines, shows showing more profits and Performs better.	It cannot determine whether news is authentic, and fake news published by competitors can impact stock prices.
23	Eunho Koo and Geonwoo Kim (2022)	LSTM, GARCH	S&P 500 index	RMSE, correlation coefficient, overlapping area	Predicting the S&P 500 index's actual volatility.	It improves the prediction performance in extreme regions (high volatility events) and can be useful for applications like financial crisis prediction.	Don't perform as effectively when the hyperparameter α is near 0.
24	Somenath et al. (2021)	ANN and CNN	NSE	accuracy	In order to forecast the stock prices for tomorrow.	CNN model requires less training data and time.	Required many training data, suffered from overfitting.
08	Zhigang et al. (2020)	CNN, EMD, and LSTM	AAPL stock	MAE, RMSE, MAPE, R ²	Stock closing price prediction.	-improve prediction accuracy. - reduce time delay.	Limited work on sentiment analysis and LSTM.
25	Abdul et al. (2023)	MLS LSTM	Yahoo Finances	R ² score RMSE, MAPE	Stock price prediction	Accuracy is higher in prediction when compared to SVM and linear regression.	Requires more time for training and resource requirements. - LSTM Overfitting

Eunho[23] used GARCH-type models, including EGARCH, TGARCH, and LSTM demonstrate that the recommended method makes the prediction's distribution resemble the target distribution. For most hyperparameter α values, the recommended VU strategy with hybrid models outperforms better than native models, except $0 < \alpha < 0.3$. Zhigang [8] proposes S_EMDAM_LSTM - a sentiment and empirical mode decomposition-based LSTM with an attention mechanism to predict the closing price of the stock market. The suggested algorithm makes three main contributions: a CNN-based sentiment index calculation module, an LSTM-based prediction module with an attention mechanism, and, an EMD-based time-series decomposition module. Abdul [26] used 4 layers in the MLS LSTM model, with three LSTM layers and one layer dense. On training data, the accuracy of the model's prediction was 95.9% and on the testing data set the accuracy achieved was 98.1%. MAPE of the MLS LSTM model was only 2.18% on the testing data, which is remarkably lesser than other deep learning architectures. RMSE is 0.019 for train data and 0.028 for test data indicating highly accurate predictions.

III. RESEARCH GAPS AND ISSUES

Despite the fact that there are many different approaches available to foresee the stock market, there are some constraints that need to be considered to achieve an accurate stock market forecast. The research gaps and problems with various stock market forecasts are covered in this section.

A major gap identified in current research is the weak of robust generalization in predictive models. On training data, many models perform exceptionally well however fail to maintain accuracy on new data due to overfitting. Among the most renowned, and overfitting is the majority challenging issue in machine learning models. Accurately predicting stock market volatility yet difficult because of the complex and unobserved nature of volatility. In some proposed models only bullish stocks showed good performance and improvement to bearish stocks as well has to be considered. Whether historical price data or sentiment analysis independently are considered by many researchers, there is limited research on integrating multimodal data sources including combining news sentiment, social media, economic indicators, and price data to improve prediction accuracy. Text data from different sources of social media platforms and news articles media data are the primary sources of sentiment analysis used in stock prediction. The possibilities of visual and auditory data (e.g., video interviews and company earnings calls) remain underexplored. Combining technical analysis and deep-learning has not received much attention for the prediction of stock prices.

IV. CONCLUSION

This study discusses the various techniques used to achieve accurate stock market predictions, categorized based on prediction methods. The aim is to classify the existing techniques based on the datasets used, performance metrics, and algorithms used. To thoroughly evaluate model performance, measures such as accuracy rates, MSE, and MAE were employed. The metrics used to measure are not unique and several writers have employed various combinations of metrics. It's interesting to note that models that included technical analysis and sentiment analysis performed better at predicting

stock price than models that just employed sentiment analysis or technical analysis. Among the models reviewed, LSTM along with neural networks demonstrated the highest performance levels. Based on the analysis, shows that a hybrid model has a better prediction accuracy as compared to the individual model.

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